

Ch. 2 — P01 (C) When to Simulate

Problem Bank | Chapter 2 | Type: C | Difficulty:

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For each system below, decide whether discrete-event simulation, an analytical model, Monte Carlo (static) simulation, ODE/system-dynamics, or no model is most appropriate. Classify each along the three axes: {Discrete, Continuous} $\times$ {Stochastic, Deterministic} $\times$ {Static, Dynamic}.

Justify each answer in 1–2 sentences.

1. A factory produces exactly 200 identical widgets per 8-hour shift. The manager wants to know the total output per week.
2. A hospital emergency department with 6 exam rooms, 3 triage nurses, and time-varying patient arrivals (high at noon, low at 3 AM). The administrator asks: “What is the 90th-percentile wait time?”
3. A financial analyst wants to estimate the probability that a stock portfolio loses more than 15% of its value in any given year, based on historical return data.
4. A water distribution network obeys Hazen-Williams flow equations. An engineer wants to find the pressure at each node.
5. An epidemiologist models flu spread through a city of 500,000 people using an SIR compartmental model ($dS/dt = -\beta SI$, etc.).
6. A call centre with 10 agents, Poisson arrivals at rate $\lambda = 20/\text{hr}$, and exponential service times with mean 4 min. The manager asks: “What is the average queue length?”
7. A parking garage with 200 spaces and Poisson arrivals/departures. Drivers leave immediately if no space is available. What fraction of drivers are turned away?
8. A single machine processes jobs with exactly 15-minute setup time and exactly 30-minute processing time, one job at a time. What is the throughput per hour?